

Air Data Box Firmware Solution Proposal

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1. Executive Summary & System Overview

Develop STM32 firmware for an Air Data Box that acquires pressure data from Honeywell RSCDJI060MDSE3, Bosch BMP581, and TE MS5611 sensors, computes airspeed and altitude, and transmits processed data over CAN.

System Flow

Pitot Probe → Pressure Sensors → STM32 → Filtering → Airspeed/Altitude Algorithms → CAN Output

Sensor Strategy

- Honeywell: Differential Pressure (airspeed source)
- BMP581: Static Pressure
- MS5611: Static Pressure Redundancy & Validation

2. Firmware Architecture

Layered Architecture

Application Layer → Service Layer → Driver Layer → MCU Layer

MCU Layer: GPIO, SPI/I2C, CAN, Timers, Interrupts

Driver Layer: Honeywell, BMP581, MS5611, CAN Drivers

Service Layer: Sensor Manager, Filters, Air Data Processing

Application Layer: Startup, Runtime Control, Fault Handling

Repository Structure

AirDataBox/

■■■ Drivers/

■■■ Services/

■■■ Application/

■■■ Docs/

■■■ Tests/

3. Sensor Driver & Sensor Management

Driver APIs

Init() | ReadPressure() | ReadTemperature() | GetStatus()

Sensor Manager Responsibilities

- Sensor Scheduling
- Data Acquisition
- Timestamping
- Validation & Health Monitoring

Validation Logic

Read Sensor → Communication Check → Range Check → Consistency Check → Accept Data

Cross Validation

BMP581 Pressure ↔ MS5611 Pressure

Difference > Threshold → Sensor Mismatch Fault

4. Signal Processing & Air Data Algorithms

Processing Chain

Validation → Filtering → Pressure Fusion → Airspeed → Altitude

Filtering

- Moving Average Filter
- First Order Low Pass Filter

Static Pressure Fusion

Static Pressure = (BMP581 + MS5611) / 2

Airspeed Equation

$V = \sqrt{2q/\rho}$

Altitude Equation

Altitude = $44330 \times (1 - (P/P_0)^{0.1903})$

Outputs

Airspeed, Altitude, Dynamic Pressure, Static Pressure, Temperature

5. Implementation Strategy

Drivers → Sensor Manager → Filtering → Air Data Algorithms → CAN Integration

Development Roadmap

1. STM32 Bring-up
2. Sensor Drivers
3. Signal Processing
4. Airspeed & Altitude Algorithms
5. CAN Integration
6. System Testing

6. Deliverables

- STM32 Firmware Architecture
- Sensor Driver Framework
- Airspeed & Altitude Processing Module
- CAN Communication Layer
- Modular Repository Structure
- Technical Documentation

7. Conclusion

The proposed solution demonstrates a structured STM32 firmware architecture with clear separation of sensor acquisition, signal processing, air-data computation, and CAN communication. The design is scalable, maintainable, and suitable for further implementation once hardware interface details are finalized.